$TmFeO_3$ Fe–Tm K^- Coupling

Restored derivation: physical J operator, PJP projection, local SU(3) qutrit basis, transported-gauge bond formulas, and the physical cross-peak mechanism (vertices, conservation laws, Liouville pathways), grounded in measured $TmFeO_3$ parameters

Working note

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Contents

1 Purpose and conceptual reset

The goal of this note is to formulate the Fe–Tm coupling in a way that is mathematically well-defined all the way down to the projected qutrit operators. The central object is the K^- coupling, namely the bilinear coupling between an Fe spin and the time-reversal-odd Tm qutrit operators

$$\boldsymbol{\lambda}_j^- = \begin{pmatrix} \lambda_j^2 \\ \lambda_j^5 \\ \lambda_j^7 \end{pmatrix}. \tag{1}$$

The superscript "−" means time-reversal odd. The notation j labels the Tm sublattice, not a spatial vector component.

Key point

There are three different mathematical objects that must not be confused:

1. $\boldsymbol{S}_i$: an Fe spin vector, stored in global crystallographic axes.
2. $\boldsymbol{J}_j$: the physical Tm angular momentum operator in the full $J = 6$ Hilbert space.
3. λ_j^a: a coordinate operator in the local three-level qutrit basis at Tm sublattice j.

Only $\boldsymbol{S}$ and the full $\boldsymbol{J}$ are axial vectors. A Gell-Mann operator λ_j^a is not a spatial vector component. It transforms by the adjoint action induced by the chosen qutrit bases.

The physically supported route is therefore

$$\boxed{\text{physical } \boldsymbol{S} \text{ and } \boldsymbol{J} \quad\longrightarrow\quad H_{\mathrm{Fe-Tm}} = \boldsymbol{S}^T A \boldsymbol{J} \quad\longrightarrow\quad P_j \boldsymbol{J} P_j \quad\longrightarrow\quad K_{ij}^- \boldsymbol{\lambda}_j^-.} \tag{2}$$

One should not start by assuming that the space-group rotation $R_x(\pi)$ acts as a closed onsite SU(3) rotation in one fixed Tm qutrit. That is precisely the dangerous shortcut.

2 Physical system: verified $TmFeO_3$ parameters

Before any formal manipulation we fix the physical setting and the numbers the model is calibrated against. All values below are experimental properties of bulk $TmFeO_3$; the simulation parameters sit close to them, and the small residual differences are stated explicitly so that "model" and "measurement" are never silently conflated.

Structure and sites. $TmFeO_3$ is an orthorhombically distorted perovskite, space group *Pbnm* (No. 62, a nonstandard setting of *Pnma*), with $Z = 4$ formula units per cell. The Fe^{3+} ions occupy the Wyckoff $4b$ sites, whose site symmetry is $\bar{1}$: *each Fe sits on an inversion centre*, so spatial inversion maps every Fe sublattice to itself. The Tm^{3+} ions occupy the $4c$ sites $(x, y, \frac{1}{4})$, whose site symmetry is the single mirror m (the plane $z = \frac{1}{4}$, i.e. m_z); inversion therefore exchanges Tm sublattices in pairs. This is exactly the Fe/Tm permutation structure used in the Pbnm-ingredients section, and it is dictated by the crystallography, not chosen. In particular, the local mirror $m_z = S_1^{-1} I$ that generates the λ^5, λ^7 sign is the physical $4c$ site mirror.

Magnetic ions. Fe^{3+} is $3d^5$, a spin-only ${}^6S_{5/2}$ ion ($S = 5/2$, $L \approx 0$, $g \approx 2$), represented classically by the unit vector $\boldsymbol{S}_i$. Tm^{3+} is $4f^{12}$ with ground multiplet ${}^3H_6$ ($J = 6$, 13 states). In the $C_s(m)$ crystal field the multiplet splits into 13 *singlets* (a non-Kramers ion in a site without rotational degeneracy has no protected doublets); the qutrit keeps the three lowest, $|E_1\rangle, |E_2\rangle, |E_3\rangle$, and represents them by local SU(3) Gell-Mann operators.

Magnetic order and the Γ_2 ground state. The Fe sublattice orders G-type antiferromagnetically at $T_N \approx 632\,\text{K}$, with a Dzyaloshinskii–Moriya canting that produces weak ferromagnetism. In Bertaut/Yamaguchi mode notation [5] the relevant collinear-mode multiplets are

$$\Gamma_2 = (F_x, C_y, G_z), \qquad \Gamma_4 = (G_x, A_y, F_z). \tag{3}$$

On cooling, $TmFeO_3$ undergoes a continuous spin reorientation $\Gamma_4 \to \Gamma_{24} \to \Gamma_2$ between $T_2 \approx 85\,\text{K}$ and $T_1 \approx 93\,\text{K}$ [3, 6]. Below $\sim 85\,\text{K}$ the equilibrium is Γ_2: the main AFM vector $\boldsymbol{G}$ lies along c (G_z) and the weak ferromagnetic moment $\boldsymbol{F}$ along a (F_x). This $F_x + G_z$ state is the "baseline" used throughout the numerical sections.

Excitation energies: the two axes of the cross peak. The two THz excitations that define the target cross peak are an Fe magnon and a Tm crystal-field transition:

Excitation	THz	meV	cm^{-1}	role
Fe quasi-FM magnon	0.28	1.16	9.3	low Fe mode
Fe quasi-AFM magnon	0.86	3.56	28.7	pumped q_{AFM} (ω_τ axis)
Tm CEF E_{12} (R_1)	0.52	2.15	17.3	detected coherence (ω_t axis)
Tm CEF E_{23} (R_2)	0.61	2.52	20.3	sets Δ_{13} with E_{12}
Tm CEF E_{13} (R_3)	1.15	4.76	38.4	third qutrit level

Conversion factor: $1\,\text{THz} = 4.136\,\text{meV} = 33.36\,\text{cm}^{-1} = 47.99\,\text{K}$. The three CEF lines are mutually consistent with a three-singlet qutrit, $E_{12} + E_{23} \approx E_{13}$ ($0.52 + 0.61 \approx 1.13\,\text{THz}$), so the onsite splittings in $H_{\text{Tm},0}$ are $\Delta_{12} \approx 0.52$ and $\Delta_{13} \approx 1.13\,\text{THz}$. The simulation uses $q_{\text{AFM}} \approx 0.90$ and $E_{12} \approx 0.50\,\text{THz}$, within a few percent of the measured 0.86 and 0.52 THz [3, 4]. The cross-peak question is whether a coherence on the $\omega_\tau = q_{\text{AFM}}$ axis can be converted into a coherence on the

$\omega_t = E_{12}$ axis; the two differ by $\Delta \approx 0.34\,\mathrm{THz}$, a number we return to under energy conservation in section 16.

3 Pbnm ingredients

We use the generators

$$S_1 : (x, y, z) \mapsto (-x, -y, z + \tfrac{1}{2}), \tag{4}$$
$$S_2 : (x, y, z) \mapsto (x + \tfrac{1}{2}, -y + \tfrac{1}{2}, -z), \tag{5}$$
$$I : (x, y, z) \mapsto (-x, -y, -z). \tag{6}$$

For axial vectors, the corresponding spin-space matrices are

$$R_{S_1} = \mathrm{diag}(-1, -1, +1) \equiv R_z, \qquad R_{S_2} = \mathrm{diag}(+1, -1, -1) \equiv R_x, \qquad R_I = \mathbf{1}. \tag{7}$$

Their product gives

$$R_{S_1 S_2} = R_y = \mathrm{diag}(-1, +1, -1). \tag{8}$$

The sublattice permutations are

Operation	Fe sublattice action	Tm sublattice action
S_1	(03)(12)	(03)(12)
S_2	(01)(23)	(01)(23)
S_1S_2	(02)(13)	(02)(13)
I	identity	(03)(12)

Thus inversion fixes each Fe sublattice but pairs $\mathrm{Tm}_0 \leftrightarrow \mathrm{Tm}_3$ and $\mathrm{Tm}_1 \leftrightarrow \mathrm{Tm}_2$.

4 Full physical Fe–Tm exchange before projection

Let a Fe–Tm bond be denoted

$$b = (i, j; \boldsymbol{n}), \tag{9}$$

where i is an Fe sublattice, j is a Tm sublattice, and $\boldsymbol{n}$ is the Tm-cell offset relative to the Fe cell. Before any qutrit truncation, the most general bilinear physical exchange is

$$\boxed{H_b^{\mathrm{phys}} = \boldsymbol{S}_i^T A_b \boldsymbol{J}_j.} \tag{10}$$

Here A_b is a real 3×3 tensor with rows labelled by Fe spin components and columns labelled by Tm angular-momentum components:

$$A_b = \begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{yx} & A_{yy} & A_{yz} \\ A_{zx} & A_{zy} & A_{zz} \end{pmatrix}_b . \tag{11}$$

Under a space-group operation g, the physical axial vectors transform as

$$\boldsymbol{S}_i \mapsto R_g \boldsymbol{S}_{\sigma_g^{\mathrm{Fe}}(i)}, \qquad \boldsymbol{J}_j \mapsto R_g \boldsymbol{J}_{\sigma_g^{\mathrm{Tm}}(j)}. \tag{12}$$

Therefore invariance of eq. (10) requires

$$\boxed{A_{gb} = R_g^T A_b R_g.} \tag{13}$$

This equation is safe because it is written in the full physical Hilbert space. No SU(3) truncation has been used.

Warning

The statement that $\boldsymbol{J}$ is an axial vector is a statement about the full $J = 6$ angular-momentum operator. It is not a statement that $(\lambda^2, \lambda^5, \lambda^7)$ form a spatial vector.

5 Local qutrit bases and the PJP projection

At each Tm sublattice j, define the local qutrit projector

$$P_j = \sum_{n=1}^{3} |E_{n,j}\rangle\langle E_{n,j}|. \tag{14}$$

The local qutrit basis is

$$\mathcal{B}_j = \{|E_{1,j}\rangle, |E_{2,j}\rangle, |E_{3,j}\rangle\}. \tag{15}$$

The Gell-Mann operator at that Tm site is defined by

$$\lambda_j^a = \sum_{m,n=1}^{3} |E_{m,j}\rangle(\lambda^a)_{mn}\langle E_{n,j}|. \tag{16}$$

Thus λ_j^a is a local operator coordinate in the Tm qutrit. It is not a component of a spatial vector.

The physical projected angular momentum is

$$\widetilde{J}_{j,\alpha} = P_j J_\alpha P_j. \tag{17}$$

In the local qutrit basis it is expanded as

$$\boxed{P_j J_\alpha P_j = \sum_{a\in\{2,5,7\}} \mu_{j,\alpha a}\lambda_j^a.} \tag{18}$$

Lemma: why only $\lambda^2, \lambda^5, \lambda^7$ appear

The restriction of (18) to the three *antisymmetric* Gell-Mann matrices is exact, not a small-multipole approximation, and it rests on one physical fact specific to this system. Tm^{3+} is a non-Kramers $4f^{12}$ ion at a site of symmetry m, so its CEF eigenstates are nondegenerate singlets and can be chosen *real* (time-reversal symmetric). In any real basis the angular-momentum operator J_α is purely imaginary and antisymmetric, hence $\langle E_m|J_\alpha|E_n\rangle$ is imaginary and antisymmetric. The only imaginary-antisymmetric Gell-Mann matrices are $\lambda^2, \lambda^5, \lambda^7$, so $P_j J_\alpha P_j$ has *no* component on $\lambda^{1,3,4,6,8}$. The time-odd (λ^-) versus time-even (λ^+) split used throughout this note is therefore a rigorous consequence of time reversal acting on a real singlet basis, not a modelling choice.

For a reference local basis, the projected moment matrix is

$$\mu_0 = \begin{pmatrix} 0 & 2.3915 & 0.9128 \\ 0 & -2.7866 & 0.4655 \\ 5.264 & 0 & 0 \end{pmatrix}, \qquad \text{columns } (2,5,7). \tag{19}$$

Equivalently,

$$P_0 J_x P_0 = 2.3915\lambda_0^5 + 0.9128\lambda_0^7, \tag{20}$$
$$P_0 J_y P_0 = -2.7866\lambda_0^5 + 0.4655\lambda_0^7, \tag{21}$$
$$P_0 J_z P_0 = 5.264\lambda_0^2. \tag{22}$$

The physical magnetic moment is

$$\boldsymbol{m}_j^{\mathrm{Tm}} = -g_J \mu_B P_j \boldsymbol{J} P_j. \tag{23}$$

Thus a lab magnetic field couples to the projected moment as

$$H_{B,\mathrm{Tm}}(t) = -g_J \mu_B \sum_{j,\alpha,a} B_\alpha(t) \mu_{j,\alpha a} \lambda_j^a. \tag{24}$$

The coefficient $\mu_{j,\alpha a}$ carries all local-basis information.

6 The actual SU(3) symmetry action: W_g and D_g

Let g map Tm sublattice j to $j' = \sigma_g^{\mathrm{Tm}}(j)$. The exact qutrit-basis map is

$$\boxed{[W_g^{(j)}]_{mn} = \langle E_{m,j'} | U_g | E_{n,j} \rangle.} \tag{25}$$

It acts on qutrit states as

$$U_g |E_{n,j}\rangle = \sum_m |E_{m,j'}\rangle [W_g^{(j)}]_{mn}. \tag{26}$$

It induces an adjoint action on qutrit operators:

$$\boxed{U_g \lambda_j^a U_g^\dagger = \sum_b \left[D_g^{(j)}\right]_{ba} \lambda_{j'}^b,} \tag{27}$$

where

$$\boxed{\left[D_g^{(j)}\right]_{ba} = \frac{1}{2} \operatorname{Tr}\left[\lambda^b W_g^{(j)} \lambda^a W_g^{(j)\dagger}\right].} \tag{28}$$

Key point

$W_g^{(j)}$ acts on qutrit states. $D_g^{(j)}$ acts on qutrit operators. R_g acts on physical spatial vector components. These are different matrices in different spaces.

The projected moment covariance is

$$U_g (P_j J_\alpha P_j) U_g^\dagger = \sum_\beta R_{g,\alpha\beta} P_{j'} J_\beta P_{j'}. \tag{29}$$

Substituting eqs. (18) and (27) gives

$$\boxed{\sum_a \mu_{j,\alpha a} \left[D_g^{(j)}\right]_{ba} = \sum_\beta R_{g,\alpha\beta} \mu_{j',\beta b}.} \tag{30}$$

In matrix notation, if D is defined by $\lambda^a \mapsto D_{ba} \lambda^b$, then

$$\boxed{\mu_{j'} = R_g^T \mu_j \, D_g^{(j)T}.} \tag{31}$$

This is the more theoretically supported relation. It replaces the unsafe shortcut $\lambda_{\mathrm{global}} = \mu^{-1} R \mu \lambda_{\mathrm{local}}$.

Warning

The matrix $\mu^{-1} R_x \mu$ is generally not the adjoint representation of a unitary qutrit rotation. The operation S_2 maps one Tm sublattice to another; it need not act as a closed onsite operation in one fixed qutrit. The remedy is to work with $W_g^{(j)}$ and $D_g^{(j)}$, or to choose transported bases that make selected $D_g^{(j)}$ trivial by construction.

7 Transported gauge: a practical but explicit convention

Choose Tm_0 as the reference qutrit. Define

$$\mathcal{B}_0 = \{|E_{1,0}\rangle, |E_{2,0}\rangle, |E_{3,0}\rangle\}, \tag{32}$$
$$\mathcal{B}_1 = U_{S_2}\mathcal{B}_0, \tag{33}$$
$$\mathcal{B}_2 = U_{S_1S_2}\mathcal{B}_0, \tag{34}$$
$$\mathcal{B}_3 = U_{S_1}\mathcal{B}_0. \tag{35}$$

This is the transported gauge. In this gauge,

$$D_{S_2}^{(0)} = D_{S_1S_2}^{(0)} = D_{S_1}^{(0)} = \mathbf{1}. \tag{36}$$

After restricting to the corresponding transported maps, the proper-coset qutrit action is therefore absorbed into the basis definition. This does not mean that R_x acts trivially on the qutrit. It means we chose the target qutrit basis to be the image of the source qutrit basis.

The physical global projected moment on site j is then

$$\boxed{P_j J_\alpha^{\mathrm{global}} P_j = \sum_{\beta,a} (R_j)_{\alpha\beta}\mu_{0,\beta a}\lambda_j^a,} \tag{37}$$

where

$$R_0 = \mathrm{diag}(+,+,+), \quad R_1 = \mathrm{diag}(+,-,-), \quad R_2 = \mathrm{diag}(-,+,-), \quad R_3 = \mathrm{diag}(-,-,+). \tag{38}$$

This is how a global lab field talks to local Tm λ_j operators in transported gauge:

$$H_{B,\mathrm{Tm}}(t) = -g_J\mu_B \sum_{j,a} \left[\sum_{\alpha,\beta} B_\alpha(t)(R_j)_{\alpha\beta}\mu_{0,\beta a}\right] \lambda_j^a. \tag{39}$$

8 Local mirror and the origin of the λ^5, λ^7 sign

Two operations map Tm_0 to Tm_3: S_1 and I. In transported gauge the Tm_3 basis is defined with S_1. Therefore the inversion map differs from the transported map by the local operation

$$h = S_1^{-1} I. \tag{40}$$

Using the coordinate maps,

$$h : (x,y,z) \mapsto (x,y,-z+\tfrac{1}{2}), \tag{41}$$

which is a local mirror m_z up to a translation.

For a J_z basis state, the mirror can be written as $m_z = IC_{2z}$. Since C_{2z} acts as $e^{-i\pi J_z}$ and the $4f^{12}$ parity is common to all states,

$$U(m_z)|J,m\rangle = (-1)^m |J,m\rangle \tag{42}$$

after dropping the common parity phase. The first two CEF states live in the even-m sector, while the third lives in the odd-m sector. Hence

$$U(m_z)|E_1\rangle = +|E_1\rangle, \qquad U(m_z)|E_2\rangle = +|E_2\rangle, \qquad U(m_z)|E_3\rangle = -|E_3\rangle. \tag{43}$$

Thus in the qutrit basis

$$W_{m_z} = \mathrm{diag}(+1, +1, -1). \tag{44}$$

Conjugating the Gell-Mann operators gives

$$\lambda^2 \sim -i|E_1\rangle\langle E_2| + i|E_2\rangle\langle E_1| \quad \Rightarrow \quad +\lambda^2, \tag{45}$$
$$\lambda^5 \sim -i|E_1\rangle\langle E_3| + i|E_3\rangle\langle E_1| \quad \Rightarrow \quad -\lambda^5, \tag{46}$$
$$\lambda^7 \sim -i|E_2\rangle\langle E_3| + i|E_3\rangle\langle E_2| \quad \Rightarrow \quad -\lambda^7. \tag{47}$$

Therefore

$$\boxed{M_- = \mathrm{diag}(+1, -1, -1) \quad \text{on} \quad (\lambda^2, \lambda^5, \lambda^7).} \tag{48}$$

9 From physical exchange to projected K^-

Project eq. (10) to the Tm qutrit:

$$\begin{aligned} H_b^{\mathrm{phys}} &= \sum_{\alpha\beta} S_{i,\alpha}(A_b)_{\alpha\beta} P_j J_\beta P_j \qquad (49)\\ &= \sum_{\alpha,a} S_{i,\alpha} \left[\sum_\beta (A_b)_{\alpha\beta}\mu_{j,\beta a}\right] \lambda_j^a. \qquad (50)\end{aligned}$$

Thus

$$\boxed{K^-_{b,\alpha a} = \sum_\beta (A_b)_{\alpha\beta}\mu_{j,\beta a}, \qquad a \in \{2,5,7\}.} \tag{51}$$

In matrix form,

$$\boxed{K_b^- = A_b \mu_j.} \tag{52}$$

The fully gauge-covariant symmetry rule follows by applying both the Fe rotation and the local qutrit adjoint action:

$$H_b^{K^-} = \boldsymbol{S}_i^T K_b^- \boldsymbol{\lambda}_j^-. \tag{53}$$

Under g,

$$\boldsymbol{S}_i \mapsto R_g \boldsymbol{S}_{\sigma_g^{\mathrm{Fe}}(i)}, \qquad \lambda_j^a \mapsto \sum_b \left[D_g^{(j)}\right]_{ba} \lambda^b_{\sigma_g^{\mathrm{Tm}}(j)}. \tag{54}$$

Therefore

$$\boxed{K^-_{gb,\beta b} = \sum_{\alpha,a} R_{g,\alpha\beta} K^-_{b,\alpha a}\left[D_g^{(j)}\right]_{ba}.} \tag{55}$$

This index equation is the safest form. In matrix notation, with the same D_{ba} convention,

$$\boxed{K_{gb}^- = R_g^T K_b^- D_g^{(j)T}.} \tag{56}$$

Implementation prescription

Equation (55) is the implementation anchor. The Fe row index is transformed by the spatial axial-vector matrix R_g. The Tm λ column index is transformed by the qutrit adjoint matrix $D_g^{(j)}$. In transported gauge many proper-coset D matrices are absorbed into the basis, but the Fe row signs remain if Fe spins are stored in global axes.

10 One K^- orbit in transported gauge

Take a reference bond with coefficient

$$K_0^- = \begin{pmatrix} k_{x2} & k_{x5} & k_{x7} \\ k_{y2} & k_{y5} & k_{y7} \\ k_{z2} & k_{z5} & k_{z7} \end{pmatrix} = \begin{pmatrix} \boldsymbol{k}_2 & \boldsymbol{k}_5 & \boldsymbol{k}_7 \end{pmatrix}. \tag{57}$$

For the direct proper-coset partners in transported gauge,

$$K_{ii}^- = R_i K_0^- . \tag{58}$$

For the inversion-related partners,

$$K_{i,3-i}^- = R_i K_0^- M_- . \tag{59}$$

Thus the orbit Hamiltonian is

$$\boxed{H_{K^-}^{\text{orbit}} = \sum_{i=0}^{3} \boldsymbol{S}_i^T R_i \left[\boldsymbol{k}_2(\lambda_i^2 + \lambda_{3-i}^2) + \boldsymbol{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \boldsymbol{k}_7(\lambda_i^7 - \lambda_{3-i}^7) \right].} \tag{60}$$

This equation is often the most useful way to read the selection rule:

$$\lambda^2 \text{ couples to the inversion-even Tm combination,} \tag{61}$$

$$\lambda^5, \lambda^7 \text{ couple to inversion-odd Tm combinations.} \tag{62}$$

It does not mean that the mirror-odd terms are identically zero. They vanish only after projecting onto an inversion-even Tm sector.

10.1 All eight coefficients of one orbit

Bond	Generating operation	Coefficient	Meaning
$(\mathrm{Fe}_0, \mathrm{Tm}_0)$	E	$K_{00}^- = K_0^-$	reference
$(\mathrm{Fe}_1, \mathrm{Tm}_1)$	S_2	$K_{11}^- = R_x K_0^-$	Fe y, z rows flip
$(\mathrm{Fe}_2, \mathrm{Tm}_2)$	$S_1 S_2$	$K_{22}^- = R_y K_0^-$	Fe x, z rows flip
$(\mathrm{Fe}_3, \mathrm{Tm}_3)$	S_1	$K_{33}^- = R_z K_0^-$	Fe x, y rows flip
$(\mathrm{Fe}_0, \mathrm{Tm}_3)$	I	$K_{03}^- = K_0^- M_-$	λ^5, λ^7 columns flip
$(\mathrm{Fe}_1, \mathrm{Tm}_2)$	$S_2 I$	$K_{12}^- = R_x K_0^- M_-$	row flip + odd columns
$(\mathrm{Fe}_2, \mathrm{Tm}_1)$	$S_1 S_2 I$	$K_{21}^- = R_y K_0^- M_-$	row flip + odd columns
$(\mathrm{Fe}_3, \mathrm{Tm}_0)$	$S_1 I$	$K_{30}^- = R_z K_0^- M_-$	row flip + odd columns

11 Fixed-Fe and fixed-Tm viewpoints

For a fixed Fe sublattice i, the mirror-odd part is

$$H_i^{5,7} = \boldsymbol{S}_i^T R_i \boldsymbol{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \boldsymbol{S}_i^T R_i \boldsymbol{k}_7(\lambda_i^7 - \lambda_{3-i}^7). \tag{63}$$

So the cancellation is in the Tm sector: if $\lambda_i^{5,7} = \lambda_{3-i}^{5,7}$, then this fixed-Fe contribution vanishes.

For a fixed Tm site j, collect the λ_j^5 terms:

$$H_{\lambda_j^5} = \left(\boldsymbol{S}_j^T R_j \boldsymbol{k}_5 - \boldsymbol{S}_{3-j}^T R_{3-j} \boldsymbol{k}_5 \right) \lambda_j^5. \tag{64}$$

Similarly,

$$H_{\lambda_j^7} = \left(\boldsymbol{S}_j^T R_j \boldsymbol{k}_7 - \boldsymbol{S}_{3-j}^T R_{3-j} \boldsymbol{k}_7 \right) \lambda_j^7. \tag{65}$$

This is not $\boldsymbol{S}_j - \boldsymbol{S}_{I(j)}$, because inversion fixes the Fe sublattice. The index $3 - j$ enters because it is the Fe partner of the other member of this particular bond orbit.

12 Including the Fe and Tm Hamiltonians

A minimal Fe Hamiltonian in global axes is

$$H_{\mathrm{Fe}} = \sum_{\langle mn\rangle_1} [J_{1xy}(S^x_m S^x_n + S^y_m S^y_n) + J_{1z} S^z_m S^z_n] \tag{66}$$

$$+ \sum_{\langle mn\rangle_2} [J_{2xy}(S^x_m S^x_n + S^y_m S^y_n) + J_{2z} S^z_m S^z_n] \tag{67}$$

$$+ \sum_m \left[D_x(S^x_m)^2 + D_y(S^y_m)^2 + D_z(S^z_m)^2\right]. \tag{68}$$

The onsite Tm qutrit Hamiltonian with $E_1 = 0$, $E_2 = \Delta_{12}$, $E_3 = \Delta_{13}$ is

$$H_{\mathrm{Tm},0} = \sum_j \left[-\frac{\Delta_{12}}{2}\lambda^3_j + \frac{\Delta_{12} - 2\Delta_{13}}{2\sqrt{3}}\lambda^8_j\right] + \text{constant}. \tag{69}$$

The full leading model restricted to K^- is

$$\boxed{H = H_{\mathrm{Fe}} + H_{\mathrm{Tm},0} + \sum_\nu H_{K^-,\nu} + H_{B,\mathrm{Fe}} + H_{B,\mathrm{Tm}} + H_{E,\mathrm{Tm}}.} \tag{70}$$

Here ν labels inequivalent Fe–Tm geometric bond orbits. Each $H_{K^-,\nu}$ has the form of eq. (60) but with its own reference columns $\boldsymbol{k}_{\nu,2}$, $\boldsymbol{k}_{\nu,5}$, $\boldsymbol{k}_{\nu,7}$ and its own cell offsets.

13 Higher-order symmetry-allowed couplings involving external fields and two Fe spins

The previous sections derived the lowest-order Fe–Tm bilinear

$$\boldsymbol{S}_i^T A_b \boldsymbol{J}_j \quad \longrightarrow \quad \boldsymbol{S}_i^T K_b^- \boldsymbol{\lambda}_j^-. \tag{71}$$

The same logic gives the higher-order couplings. The key rule is that one must first write a physical operator with well-defined spatial and time-reversal transformation properties, and only then project the Tm operator into the local SU(3) qutrit space.

We will use two Tm qutrit sectors:

$$\boldsymbol{\lambda}_j^- = (\lambda^2_j, \lambda^5_j, \lambda^7_j)^T, \qquad \text{time-reversal odd,} \tag{72}$$

$$\boldsymbol{\lambda}_j^+ = (\lambda^1_j, \lambda^3_j, \lambda^4_j, \lambda^6_j, \lambda^8_j)^T, \qquad \text{time-reversal even.} \tag{73}$$

Often only the off-diagonal time-even coherences are kept,

$$\boldsymbol{\lambda}^+_{j,\mathrm{off}} = (\lambda^1_j, \lambda^4_j, \lambda^6_j)^T. \tag{74}$$

The local mirror m_z acts on the three CEF states as

$$W_{m_z} = \mathrm{diag}(+1, +1, -1). \tag{75}$$

Therefore the mirror/parity matrices in the local qutrit basis are

$$M_- = \mathrm{diag}(+1, -1, -1), \qquad \text{basis } (\lambda^2, \lambda^5, \lambda^7), \tag{76}$$

$$M_+ = \mathrm{diag}(+1, +1, -1, -1, +1), \qquad \text{basis } (\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8), \tag{77}$$

$$M_{+,\mathrm{off}} = \mathrm{diag}(+1, -1, -1), \qquad \text{basis } (\lambda^1, \lambda^4, \lambda^6). \tag{78}$$

Key point

Time reversal fixes which qutrit sector can appear:

Factor	$\mathcal{T}$ parity	Inversion type	Comment
$\boldsymbol{S}$	odd	axial, even under I	Fe spin
$\boldsymbol{J}$	odd	axial, even under I	physical Tm angular momentum
$\boldsymbol{B}$	odd	axial, even under I	magnetic field
$\boldsymbol{E}$	even	polar, odd under I	electric field
λ^-	odd	M_-	$\lambda^2, \lambda^5, \lambda^7$
λ^+	even	M_+	$\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8$

A term in the Hamiltonian must be time-reversal even unless the external nonequilibrium drive is being treated as an explicitly time-dependent source. Thus ESJ uses the λ^- sector after PJP projection, while $BS\lambda$ and $SS\lambda$ naturally use the λ^+ sector.

13.1 Electric-field-assisted Fe–Tm term: $E + S + J$

Start from a physical third-order coupling on a bond $b = (i, j; \boldsymbol{n})$,

$$\boxed{H_b^{ESJ}(t) = \sum_{\eta,\alpha,\beta} E_\eta(t)\, S_{i,\alpha}\, C^{EJ}_{b;\eta\alpha\beta}\, J_{j,\beta}.} \tag{79}$$

Here η is a polar-vector electric-field index, α is an Fe-spin axial-vector index, and β is a physical Tm angular-momentum axial-vector index. Since E is time-even while S and J are time-odd, the product ESJ is time-even.

Now project the physical J operator:

$$P_j J_\beta P_j = \sum_{a\in\{2,5,7\}} \mu_{j,\beta a} \lambda_j^a. \tag{80}$$

Then

$$P_j H_b^{ESJ} P_j = \sum_{\eta,\alpha,a\in\{2,5,7\}} E_\eta(t) S_{i,\alpha} K^{E,-}_{b;\eta\alpha a} \lambda_j^a, \tag{81}$$

with

$$\boxed{K^{E,-}_{b;\eta\alpha a} = \sum_\beta C^{EJ}_{b;\eta\alpha\beta} \mu_{j,\beta a}.} \tag{82}$$

Thus $E + S + J$ is not a new kind of Tm operator; after projection it is an electric-field-assisted source for the same time-odd qutrit operators $\lambda^2, \lambda^5, \lambda^7$.

Gauge-covariant symmetry rule. For a space-group operation g, the polar electric field transforms with the polar matrix Q_g, while S and J transform with the axial matrix R_g. In the projected qutrit basis,

$$U_g \lambda_j^a U_g^\dagger = \sum_b \left[D_g^{(j)}\right]_{ba} \lambda^b_{\sigma_g(j)}. \tag{83}$$

Therefore the projected tensor satisfies the index rule

$$\boxed{K^{E,-}_{gb;\nu\gamma c} = \sum_{\eta,\alpha,a} Q_{g,\eta\nu}\, R_{g,\alpha\gamma}\, K^{E,-}_{b;\eta\alpha a} \left[D_g^{(j)}\right]_{ca}.} \tag{84}$$

This is the field-assisted analogue of the K^- covariance rule. The extra Q_g index is the only new ingredient.

Inversion partner. For inversion, $Q_I = -\mathbf{1}$ because E is polar, while $R_I = +\mathbf{1}$ for axial vectors. In the transported gauge the relative Tm operation gives M_-, so the inversion partner carries

$$\boxed{\text{odd-bond sign matrix for } ESJ: \quad -M_- = \mathrm{diag}(-1,+1,+1).} \tag{85}$$

Thus the selection rule is the opposite of the static K^- term:

$$\lambda^2 : \text{opposite sign on the inversion partner,} \tag{86}$$
$$\lambda^5, \lambda^7 : \text{same sign on the inversion partner.} \tag{87}$$

Transported-gauge orbit formula. Let the reference electric-assisted column vectors be

$$\boldsymbol{e}_{\eta,2}, \quad \boldsymbol{e}_{\eta,5}, \quad \boldsymbol{e}_{\eta,7}, \tag{88}$$

where each is a three-component Fe-spin vector. For the proper coset sending the reference Fe sublattice to i, define Q_i as the polar point matrix and R_i as the axial spin matrix. A compact transported-gauge form is

$$\boxed{\begin{aligned} H^{ESJ}_{\text{orbit}}(t) = \sum_{i=0}^{3}\sum_{\nu} E_\nu(t)\, \boldsymbol{S}_i^T R_i \sum_{\eta} Q_{i,\eta\nu} \Big[& \boldsymbol{e}_{\eta,2}\big(\lambda_i^2 - \lambda_{3-i}^2\big) \\ & + \boldsymbol{e}_{\eta,5}\big(\lambda_i^5 + \lambda_{3-i}^5\big) + \boldsymbol{e}_{\eta,7}\big(\lambda_i^7 + \lambda_{3-i}^7\big)\Big]. \end{aligned}} \tag{89}$$

If the code has already contracted the electric tensor onto a fixed lab polarization, the factor $\sum_\eta Q_{i,\eta\nu} E_\nu$ should be viewed as a sublattice-dependent effective coefficient. Dropping it is only valid if the coupling parameters have already been fitted separately for each proper-coset partner.

Implementation prescription

For an implementation storing Fe spins in global axes and Tm operators in local qutrit bases, the ESJ odd-bond signs are

$$\lambda^2 : -1, \qquad \lambda^5 : +1, \qquad \lambda^7 : +1. \tag{90}$$

Additionally, proper-coset partners require the Fe row signs R_i, and field-assisted terms require the electric-field polar-index signs Q_i unless the field polarization has already been folded into orbit-dependent coefficients.

13.2 Magnetic-field-assisted Fe–Tm term: $B + S + \lambda$

A magnetic-field-assisted coupling with one Fe spin and one Tm qutrit operator has the form

$$\boxed{H_b^{BS\lambda}(t) = \sum_{\eta,\alpha,a\in+} B_\eta(t)\, S_{i,\alpha}\, K^{B,+}_{b;\eta\alpha a}\, \lambda_j^a.} \tag{91}$$

Here $a \in +$ means

$$a \in \{1,3,4,6,8\}, \tag{92}$$

or, if only off-diagonal coherences are retained,

$$a \in \{1,4,6\}. \tag{93}$$

This time-even sector is required because B and S are both time-reversal odd, so BS is time-reversal even.

If one wants to derive this term from a physical Tm time-even operator $\mathcal{O}_\rho^+$, write

$$P_j \mathcal{O}_\rho^+ P_j = o_{j,\rho 0}\mathbf{1} + \sum_{a\in\{1,3,4,6,8\}} o_{j,\rho a}\lambda_j^a. \tag{94}$$

A physical coupling

$$\sum_{\eta,\alpha,\rho} B_\eta S_{i,\alpha} L_{b;\eta\alpha\rho}\mathcal{O}_{j,\rho}^+ \tag{95}$$

projects to eq. (91) with

$$\boxed{K_{b;\eta\alpha a}^{B,+} = \sum_\rho L_{b;\eta\alpha\rho} o_{j,\rho a}.} \tag{96}$$

This is the even-sector analogue of the PJP step.

Gauge-covariant symmetry rule. The magnetic field is an axial vector, so it transforms with R_g, not Q_g. Therefore

$$\boxed{K_{gb;\nu\gamma c}^{B,+} = \sum_{\eta,\alpha,a} R_{g,\eta\nu}\, R_{g,\alpha\gamma}\, K_{b;\eta\alpha a}^{B,+} \left[D_{g,+}^{(j)}\right]_{ca}.} \tag{97}$$

Here $D_{g,+}$ is the adjoint action restricted to the time-even qutrit sector.

Inversion partner. Under inversion, B is axial and hence does not change sign. S is also axial. Thus the odd-bond sign is just the time-even qutrit mirror matrix:

$$\boxed{\text{odd-bond sign matrix for } BS\lambda^+ : \quad M_+ = \operatorname{diag}(+1,+1,-1,-1,+1).} \tag{98}$$

For the off-diagonal subset $(\lambda^1, \lambda^4, \lambda^6)$,

$$\boxed{M_{+,\text{off}} = \operatorname{diag}(+1,-1,-1).} \tag{99}$$

So

$$\lambda^1, \lambda^3, \lambda^8 : \text{same sign on the inversion partner,} \tag{100}$$

$$\lambda^4, \lambda^6 : \text{opposite sign on the inversion partner.} \tag{101}$$

Transported-gauge orbit formula, off-diagonal version. For reference column vectors $\boldsymbol{b}_{\eta,1}$, $\boldsymbol{b}_{\eta,4}$ and $\boldsymbol{b}_{\eta,6}$,

$$\boxed{\begin{aligned} H_{\text{orbit,off}}^{BS\lambda}(t) = \sum_{i=0}^{3}\sum_\nu B_\nu(t)\, \boldsymbol{S}_i^T R_i \sum_\eta R_{i,\eta\nu}\Big[&\boldsymbol{b}_{\eta,1}(\lambda_i^1 + \lambda_{3-i}^1) \\ &+ \boldsymbol{b}_{\eta,4}(\lambda_i^4 - \lambda_{3-i}^4) + \boldsymbol{b}_{\eta,6}(\lambda_i^6 - \lambda_{3-i}^6)\Big]. \end{aligned}} \tag{102}$$

If the diagonal even operators are included, add

$$\boldsymbol{b}_{\eta,3}(\lambda_i^3 + \lambda_{3-i}^3) + \boldsymbol{b}_{\eta,8}(\lambda_i^8 + \lambda_{3-i}^8). \tag{103}$$

to the bracket.

Implementation prescription

For $BS\lambda^+$, the inversion signs are not the same as the static K^- term. They are

$$\lambda^1 : +1, \qquad \lambda^4 : -1, \qquad \lambda^6 : -1 \tag{104}$$

for the off-diagonal sector. Proper-coset partners require Fe row signs R_i and magnetic-field axial-index signs R_i.

13.3 Fe–Fe–Tm term: $S + S + \lambda$

A symmetry-allowed term with two Fe spins and one Tm qutrit operator is

$$\boxed{H_b^{SS\lambda} = \sum_{m,n} \sum_{\alpha,\beta,a\in+} S_{m,\alpha}\, S_{n,\beta}\, W_{b;mn;\alpha\beta a}\, \lambda_j^a.} \tag{105}$$

Because two Fe spins are time-reversal even as a product, the Tm operator must again be time-reversal even:

$$a \in \{1, 3, 4, 6, 8\}. \tag{106}$$

For the onsite version used in many minimal simulations, $m = n = i$ and

$$\boxed{H_{b,\text{onsite}}^{SS\lambda} = \sum_{a\in+} \boldsymbol{S}_i^T W_b^a \boldsymbol{S}_i\, \lambda_j^a.} \tag{107}$$

For a single classical spin on the same site, only the symmetric part of W^a matters, so one may parameterize

$$W^a = \begin{pmatrix} W_{xx}^a & W_{xy}^a & W_{xz}^a \\ W_{xy}^a & W_{yy}^a & W_{yz}^a \\ W_{xz}^a & W_{yz}^a & W_{zz}^a \end{pmatrix}. \tag{108}$$

This is the origin of the six real parameters (xx, yy, zz, xy, xz, yz) per λ^a channel in a code implementation.

If the term is derived from a physical time-even Tm multipole $\mathcal{O}_\rho^+$, then

$$P_j \mathcal{O}_\rho^+ P_j = o_{j,\rho 0}\mathbf{1} + \sum_{a\in+} o_{j,\rho a}\lambda_j^a, \tag{109}$$

and the physical coupling

$$\sum_{\alpha,\beta,\rho} S_{m,\alpha} S_{n,\beta} \mathcal{W}_{b;mn;\alpha\beta\rho} \mathcal{O}_{j,\rho}^+ \tag{110}$$

projects to eq. (105) with

$$\boxed{W_{b;mn;\alpha\beta a} = \sum_\rho \mathcal{W}_{b;mn;\alpha\beta\rho} o_{j,\rho a}.} \tag{111}$$

Gauge-covariant symmetry rule. For the general two-Fe-site case,

$$\boxed{W_{gb;\gamma\delta c} = \sum_{\alpha,\beta,a} R_{g,\alpha\gamma}\, R_{g,\beta\delta}\, W_{b;\alpha\beta a} \left[D_{g,+}^{(j)}\right]_{ca}.} \tag{112}$$

For the onsite matrix form $\boldsymbol{S}^T W^a \boldsymbol{S}$, this becomes

$$\boxed{W_{gb}^c = \sum_a \left[D_{g,+}^{(j)}\right]_{ca} R_g^T W_b^a R_g.} \tag{113}$$

Since the R_i used here are diagonal π-rotation matrices, $R_i^T = R_i$.

Inversion partner. There is no polar electric prefactor and no odd physical J operator. Hence the inversion sign is again just the time-even qutrit parity matrix M_+:

$$\boxed{\text{odd-bond sign matrix for } SS\lambda^+ : \quad M_+ = \mathrm{diag}(+1, +1, -1, -1, +1).} \tag{114}$$

Thus $\lambda^1, \lambda^3, \lambda^8$ couple to inversion-even Tm combinations, whereas λ^4, λ^6 couple to inversion-odd Tm combinations.

Transported-gauge onsite orbit formula. Keeping only the off-diagonal even coherences for compactness,

$$\boxed{\begin{aligned} H^{SS\lambda}_{\text{orbit,off}} = \sum_{i=0}^{3} \Big[& \boldsymbol{S}_i^T R_i W^1 R_i \boldsymbol{S}_i \, (\lambda_i^1 + \lambda_{3-i}^1) \\ & + \boldsymbol{S}_i^T R_i W^4 R_i \boldsymbol{S}_i \, (\lambda_i^4 - \lambda_{3-i}^4) \\ & + \boldsymbol{S}_i^T R_i W^6 R_i \boldsymbol{S}_i \, (\lambda_i^6 - \lambda_{3-i}^6) \Big]. \end{aligned}} \tag{115}$$

If λ^3 and λ^8 are retained, add

$$\boldsymbol{S}_i^T R_i W^3 R_i \boldsymbol{S}_i (\lambda_i^3 + \lambda_{3-i}^3) + \boldsymbol{S}_i^T R_i W^8 R_i \boldsymbol{S}_i (\lambda_i^8 + \lambda_{3-i}^8). \tag{116}$$

13.4 Sign-summary table for implementation

For one transported-gauge inversion pair, the relative sign on the odd bond is the product of three contributions:

$$\boxed{\text{odd-bond sign} = (\text{external-field inversion parity}) \times (\text{local qutrit mirror parity}).} \tag{117}$$

Fe spin is axial, so it does not add an inversion sign. However, proper cosets still require R_i row signs for Fe components if Fe spins are stored in global axes.

Term	Tm sector	Odd-bond sign matrix	Surviving inversion-even Tm combination
$SJ \to S\lambda^-$	$(2, 5, 7)$	$M_- = (+, -, -)$	λ^2
$ESJ \to ES\lambda^-$	$(2, 5, 7)$	$-M_- = (-, +, +)$	λ^5, λ^7
$BS\lambda^+_{\text{off}}$	$(1, 4, 6)$	$M_{+,\text{off}} = (+, -, -)$	λ^1
$SS\lambda^+_{\text{off}}$	$(1, 4, 6)$	$M_{+,\text{off}} = (+, -, -)$	λ^1
$BS\lambda^+$ or $SS\lambda^+$	$(1, 3, 4, 6, 8)$	$(+, +, -, -, +)$	$\lambda^1, \lambda^3, \lambda^8$

Warning

The phrase "surviving inversion-even Tm combination" refers only to the pair sum/ difference induced by the inversion partner. It does not say that the full response is nonzero for every Fe magnetic pattern or every field polarization. Proper-coset row signs, field-index signs, and the actual Fe mode pattern can impose additional cancellations.

13.5 Complete Hamiltonian with every term written out

Collecting the Fe sector, the onsite Tm sector, the leading bilinear K^-, the three higher-order field/two-spin vertices, and the external-field drives, the full transported-gauge model is

$$\boxed{H(t) = H_{\text{Fe}} + H_{\text{Tm},0} + \sum_\nu \Big(H_\nu^{K^-} + H_\nu^{ESJ}(t) + H_\nu^{BS\lambda}(t) + H_\nu^{SS\lambda} \Big) + H_{B,\text{Fe}}(t) + H_{B,\text{Tm}}(t) + H_{E,\text{Tm}}(t),} \tag{118}$$

where ν runs over inequivalent Fe–Tm bond orbits and $i = 0, 1, 2, 3$ over the four sublattices, with the proper-coset row matrices

$$R_0 = \mathrm{diag}(+,+,+), \quad R_1 = \mathrm{diag}(+,-,-), \quad R_2 = \mathrm{diag}(-,+,-), \quad R_3 = \mathrm{diag}(-,-,+). \tag{119}$$

Every term is now written explicitly.

Fe sector (first/second neighbour exchange and single-ion anisotropy).

$$\begin{aligned} H_{\mathrm{Fe}} = &\sum_{\langle mn\rangle_1} \big[J_{1xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{1z} S_m^z S_n^z\big] + \sum_{\langle mn\rangle_2} \big[J_{2xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{2z} S_m^z S_n^z\big] \\ &+ \sum_m \big[D_x (S_m^x)^2 + D_y (S_m^y)^2 + D_z (S_m^z)^2\big]. \end{aligned} \tag{120}$$

Onsite Tm sector (three-singlet qutrit).

$$H_{\mathrm{Tm},0} = \sum_j \Big[-\tfrac{\Delta_{12}}{2}\,\lambda_j^3 + \tfrac{\Delta_{12}-2\Delta_{13}}{2\sqrt{3}}\,\lambda_j^8 \Big], \qquad \Delta_{12} \approx 0.52\,\mathrm{THz}, \quad \Delta_{13} \approx 1.13\,\mathrm{THz}. \tag{121}$$

Static bilinear K^- (columns $\boldsymbol{k}_{\nu,2}, \boldsymbol{k}_{\nu,5}, \boldsymbol{k}_{\nu,7}$), λ^- sector.

$$H_\nu^{K^-} = \sum_{i=0}^{3} \boldsymbol{S}_i^T R_i \big[\boldsymbol{k}_{\nu,2}(\lambda_i^2 + \lambda_{3-i}^2) + \boldsymbol{k}_{\nu,5}(\lambda_i^5 - \lambda_{3-i}^5) + \boldsymbol{k}_{\nu,7}(\lambda_i^7 - \lambda_{3-i}^7)\big]. \tag{122}$$

Electric-field-assisted ESJ (columns $\boldsymbol{e}_{\eta,a}$), λ^- sector.

$$H_\nu^{ESJ}(t) = \sum_{i=0}^{3}\sum_\eta E_\eta(t)\, \boldsymbol{S}_i^T R_i \big[\boldsymbol{e}_{\eta,2}(\lambda_i^2 - \lambda_{3-i}^2) + \boldsymbol{e}_{\eta,5}(\lambda_i^5 + \lambda_{3-i}^5) + \boldsymbol{e}_{\eta,7}(\lambda_i^7 + \lambda_{3-i}^7)\big], \tag{123}$$

where the polar field carries $\boldsymbol{e}_{\eta,a} \to \sum_\kappa Q_{i,\eta\kappa} \boldsymbol{e}_{\kappa,a}$ with $Q_I = -\mathbf{1}$ (this is the origin of the relative $-M_-$ versus K^-).

Magnetic-field-assisted $BS\lambda$ (columns $\boldsymbol{b}_{\eta,a}$), λ^+ sector.

$$\begin{aligned} H_\nu^{BS\lambda}(t) = \sum_{i=0}^{3}\sum_\eta B_\eta(t)\, \boldsymbol{S}_i^T R_i \Big[&\boldsymbol{b}_{\eta,1}(\lambda_i^1 + \lambda_{3-i}^1) + \boldsymbol{b}_{\eta,4}(\lambda_i^4 - \lambda_{3-i}^4) + \boldsymbol{b}_{\eta,6}(\lambda_i^6 - \lambda_{3-i}^6) \\ &+ \boldsymbol{b}_{\eta,3}(\lambda_i^3 + \lambda_{3-i}^3) + \boldsymbol{b}_{\eta,8}(\lambda_i^8 + \lambda_{3-i}^8)\Big], \end{aligned} \tag{124}$$

with $\boldsymbol{b}_{\eta,a} \to \sum_\kappa R_{i,\eta\kappa} \boldsymbol{b}_{\kappa,a}$ (axial field, hence R_i not Q_i).

Two-Fe-spin $SS\lambda$ (onsite symmetric matrices W^a), λ^+ sector.

$$\begin{aligned} H_\nu^{SS\lambda} = \sum_{i=0}^{3} \Big[&(\boldsymbol{S}_i^T R_i W^1 R_i \boldsymbol{S}_i)(\lambda_i^1 + \lambda_{3-i}^1) + (\boldsymbol{S}_i^T R_i W^4 R_i \boldsymbol{S}_i)(\lambda_i^4 - \lambda_{3-i}^4) + (\boldsymbol{S}_i^T R_i W^6 R_i \boldsymbol{S}_i)(\lambda_i^6 - \lambda_{3-i}^6) \\ &+ (\boldsymbol{S}_i^T R_i W^3 R_i \boldsymbol{S}_i)(\lambda_i^3 + \lambda_{3-i}^3) + (\boldsymbol{S}_i^T R_i W^8 R_i \boldsymbol{S}_i)(\lambda_i^8 + \lambda_{3-i}^8)\Big]. \end{aligned} \tag{125}$$

External-field drives.

$$H_{B,\mathrm{Fe}}(t) = g_{\mathrm{Fe}}\mu_B \sum_m \boldsymbol{B}(t)\cdot \boldsymbol{S}_m, \tag{126}$$

$$H_{B,\mathrm{Tm}}(t) = -g_J\mu_B \sum_j \sum_{a\in\{2,5,7\}} \Big[\sum_{\alpha,\beta} B_\alpha(t)\,(R_j)_{\alpha\beta}\,\mu_{0,\beta a}\Big]\lambda_j^a, \tag{127}$$

$$H_{E,\mathrm{Tm}}(t) = \sum_j \sum_{a\in\{1,3,4,6,8\}} \Big[\sum_\eta E_\eta(t)\,p_{j,\eta a}\Big]\lambda_j^a, \tag{128}$$

where μ_0 is the projected-moment matrix (19) and $p_{j,\eta a}$ is the projected Tm electric-multipole tensor (time-even, hence λ^+).

Time-reversal sector bookkeeping. The λ^- sector $\{2,5,7\}$ is fed by K^- and ESJ; the λ^+ sector $\{1,3,4,6,8\}$ is fed by $BS\lambda$, $SS\lambda$, and $H_{E,\mathrm{Tm}}$. $H_{\mathrm{Fe}}, H_{\mathrm{Tm},0}, H^{K^-}, H^{SS\lambda}, H_{B,\mathrm{Fe}}, H_{B,\mathrm{Tm}}$ are time-even by construction; the field-assisted terms are time-even only together with their explicit external field. This sorting is exactly what the cross-peak analysis of section 16 acts on.

14 Implementation remedy

14.1 If Fe spins are stored in global axes

Use the transported-gauge local λ_j basis for Tm. Then for each bond in a symmetry orbit:

$$\boxed{K^-_{\mathrm{bond}} = R_{\mathrm{Fe(bond)}} K^-_\nu M_{\mathrm{inv}},} \tag{129}$$

where

$$M_{\mathrm{inv}} = \begin{cases} \mathbf{1}, & \text{direct/proper-coset branch,} \\ M_- = \mathrm{diag}(+1,-1,-1), & \text{inversion-related branch.} \end{cases} \tag{130}$$

This is the version that matches a code convention in which Fe spins are lab Cartesian and Tm SU(3) states are local CEF-basis variables.

14.2 If Fe spins are instead stored in sublattice local frames

Then the R_i row signs may be absorbed into the spin variables themselves. In that convention the bond tensor should not multiply by R_i again. The code and the theory must choose one convention and use it consistently.

Warning

Do not mix these conventions. If Fe spins are global, include the R_i row signs in the bond tensor. If Fe spins are local, do not include them again. The Tm λ_j operators remain local qutrit coordinates either way.

14.3 Do not use $\mu^{-1}R\mu$ as a qutrit frame by default

The safe object is $P_j J_\alpha P_j$ or, in local coordinates, $\mu_{j,\alpha a}\lambda_j^a$. If a global observable is needed, compute it through

$$J^{\mathrm{global}}_{\alpha,j} = \sum_{\beta,a} (R_j)_{\alpha\beta}\mu_{0,\beta a}\lambda_j^a \tag{131}$$

in transported gauge, or through the fully general W_g, D_g construction. Avoid treating

$$\mu^{-1}R\mu \tag{132}$$

as a genuine SU(3) adjoint representation unless it has been verified to be a proper orthogonal adjoint action induced by a unitary W.

15 Diagrammatic summary

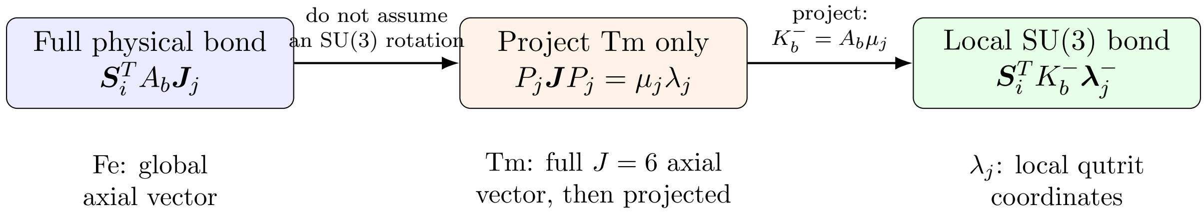

Figure 1: The correct order of operations. The Tm angular momentum is projected only after the physical symmetry constraints are formulated.

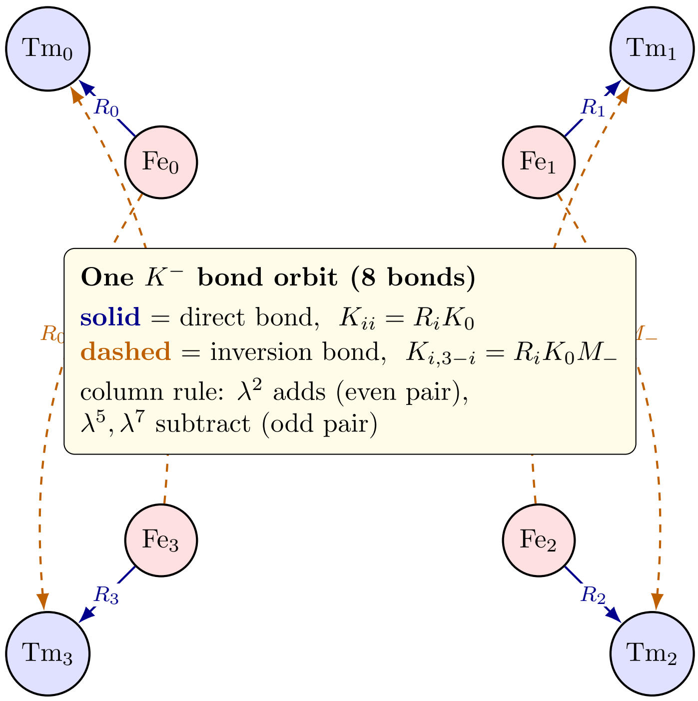

Figure 2: One transported-gauge K^- orbit. Fe sites (red) sit on inversion centres; Tm sites (blue) are exchanged in pairs by inversion ($\mathrm{Tm}_0 \leftrightarrow \mathrm{Tm}_3$, $\mathrm{Tm}_1 \leftrightarrow \mathrm{Tm}_2$). Each Fe participates in one *direct* bond (solid, coefficient $R_i K_0$) to its own Tm and one *inversion* bond (dashed, coefficient $R_i K_0 M_-$) to the partner Tm. Summing the orbit gives the even/odd column rule of (60).

16 Modeling the 2DCS cross peak: what must a vertex do?

The earlier sections fixed the *form* of every symmetry-allowed Fe–Tm coupling. We now ask the sharp dynamical question: which of them can actually produce the measured $(q_{\mathrm{AFM}}, E_{12})$ 2DCS cross peak? The strategy is two-step. First we reduce the whole experiment to a *single* sharp requirement on an effective Tm force (this section). Then we test each candidate coupling against it as a Feynman-diagram catalogue (section 17). The experimental context is third-order magnetic-dipole THz 2DCS of orthoferrite magnons [1]; the language is standard nonlinear response theory [2].

16.1 The observable and $M_{\rm NL}$, precisely

Two collinear THz pulses give the total drive $B(t) = B_A(t+\tau)+B_B(t)$. The detected magnetization is a functional of that field; expand it in powers of the drive,

$$M[B] = \sum_{n\geq 1} M^{(n)}[\underbrace{B,\ldots,B}_{n}]. \tag{133}$$

With $M_A \equiv M[B_A]$, $M_B \equiv M[B_B]$, $M_{AB} \equiv M[B_A + B_B]$, the isolated nonlinear signal is

$$\boxed{M_{\rm NL}(t,\tau) = M_{AB} - M_A - M_B\,.} \tag{134}$$

Order by order, $M^{(n)}[B_A + B_B] - M^{(n)}[B_A] - M^{(n)}[B_B]$ keeps only the *mixed* terms, those with at least one factor of each pulse:

$$n = 1: \quad 0 \quad \text{(linear, no mixed term)}, \tag{135}$$

$$n = 2: \quad 2\,M^{(2)}[B_A, B_B] \quad (\chi^{(2)}\text{-type}, \ \propto B_A B_B), \tag{136}$$

$$n = 3: \quad M^{(3)} \text{ mixed} \quad (\chi^{(3)}\text{-type}, \ B_A^2 B_B + B_A B_B^2). \tag{137}$$

So $M_{\rm NL}$ is purely nonlinear and starts at second order; the $n = 1$ piece cancels identically. For magnetic-dipole driving in a centrosymmetric magnet the leading robust contribution is third order [1]. The detected quantity at the target is the Tm c-axis dipole, $m_z^{\rm Tm} \propto \lambda^2$, so the object of interest is $M_{\rm NL}^{\lambda^2}(t,\tau)$, whose 2D transform should peak at $(\omega_\tau, \omega_t) = (q_{\rm AFM}, E_{12})$.

16.2 The detection channel: only λ^2 is bright

In the E_{12} block the coherence is carried by $\lambda^1 = \mathrm{Re}|1\rangle\langle 2|$ and $\lambda^2 = \mathrm{Im}|1\rangle\langle 2|$, both oscillating at $\omega_{12} = E_{12}$, while λ^3 is the population imbalance. From (19) the radiated moment is

$$m_z^{\rm Tm} \propto P_j J_z P_j = 5.264\,\lambda^2, \tag{138}$$

so *only* λ^2 *is magnetically bright*. The onsite Hamiltonian $H_{{\rm Tm},0} \supset -\frac{\Delta_{12}}{2}\lambda^3$ generates, through $[\lambda^3, \lambda^1] = 2i\lambda^2$, the planar precession

$$\dot\lambda^1 = -\Delta_{12}\,\lambda^2, \qquad \dot\lambda^2 = +\Delta_{12}\,\lambda^1. \tag{139}$$

A force on *either* λ^1 or λ^2 therefore launches the same ring at ω_{12}. We may treat the Tm E_{12} sector as a single driven oscillator,

$$\lambda^2(t) = \int_0^t \sin\bigl[\omega_{12}(t-t')\bigr]\, h^1(t')\, dt', \tag{140}$$

driven by the effective transverse field $h^1(t)$ that a conversion vertex exerts on λ^1.

16.3 Reframing to a sharp question

Equation (140) factorizes the 2D map. The *detection*-axis (t) structure is fixed entirely by the Tm pole at E_{12}: an impulsive step force gives a $1 - \cos\omega_{12}t$ ring, a delta kick gives a $\sin\omega_{12}t$ ring. All the *delay*-axis (τ) information—in particular whether the peak sits at $\omega_\tau = q_{\rm AFM}$—lives in the τ-dependence of $h^1(t,\tau)$. Hence the entire experiment collapses to one question:

The sharp question

A coupling produces the $(q_{\rm AFM}, E_{12})$ cross peak *iff* the effective Tm force it generates,

$$M_{\rm NL}^{\lambda^2}(\tau, t) \;\propto\; f(\omega_q \tau)\,\bigl[1 - \cos\omega_{12}t\bigr], \tag{141}$$

has f *oscillating at* $\omega_q = q_{\rm AFM}$. Equivalently, the force $h^1(t,\tau)$ on the bright λ^1 channel must be (i) nonzero by symmetry, (ii) able to reach λ^1/λ^2, and (iii) carry an ω_q modulation along the delay τ.

Everything below is the evaluation of $h^1(t,\tau)$ for each candidate vertex.

16.4 Feynman rules: fields, charges, and the vertex requirement

We treat the problem as a tree-level field theory. The propagating "fields" (internal/external lines) are

line	object	properties
Fe magnon	δS_α	pole at ω_q; time-odd; the pump excites δS_y
Tm qutrit	λ^a	poles at CEF gaps; $\lambda^- = \{2,5,7\}$ (T-odd), $\lambda^+ = \{1,3,4,6,8\}$ (T-even)
B-photon	$B_\eta(t)$	external; time-odd, axial
E-photon	$E_\eta(t)$	external; time-even, polar

Every Fe magnon line is born from a pulse through the Zeeman source $g_{\rm Fe}\mu_B\, \boldsymbol{B}\cdot\boldsymbol{S}$, so each external Fe leg costs one factor of the drive. Three "charges" are conserved at every vertex:

charge	rule at a vertex
energy ω	a static vertex conserves it $\Rightarrow$ connects only equal frequencies; an external leg injects $\hbar\omega_{\rm drive}$
$\mathcal{T}$-parity	each term is time-even: λ^- needs an *odd* number of T-odd legs (S, B); λ^+ an *even* number
irrep Γ	$\Gamma(\text{legs}) \otimes \Gamma(\lambda) \supset \Gamma_1$; the orbit sign sums of (60) are this projector

Detection and the $M_{\rm NL}$ subtraction add two more bookkeeping rules: the vertex must reach λ^1/λ^2, and the diagram must attach at least one leg to each pulse. Collecting, a successful cross-peak vertex must pass all six tests:

Scorecard R1–R6

R1 (detection) reaches λ^1/λ^2 (the E_{12} block).

R2 (drive) is sourced by the pumped δS_y magnon.

R3 (parity) is $\mathcal{T}$-even, which fixes the $\lambda^\pm$ sector.

R4 (irrep) is Γ_1-allowed for the $\Gamma_2/q_{\rm AFM}$ geometry.

R5 (energy) can carry the $0.9 \to 0.5$ THz mismatch—needs ≥ 2 non-λ legs (two magnons, or one magnon + one photon).

R6 ($M_{\rm NL}$) involves both pulses.

17 The vertex catalogue: a Feynman diagram for each coupling

We now write each candidate as an interaction vertex—a term in H with a coupling constant and a fixed set of legs—draw its tree diagram for the 2DCS process, and score it against R1–R6. In every diagram time runs left to right: a red solid line is an Fe magnon (ω_q), a blue dashed line a Tm coherence, a wavy arrow an external-field interaction (one pulse); a filled dot is a field-gated bilinear vertex, a filled square a two-spin vertex. The boxed expression above each diagram is the vertex itself.

17.1 K^-: the static bilinear (fails R3/R4/R5)

$$\boxed{H_{K^-} = K^-_{a\alpha}\, S_\alpha\, \lambda^a, \qquad a \in \{2,5,7\}} \qquad [\,1\ \text{Fe (odd)} + 1\ \text{Tm}^-\text{(odd)};\ \mathcal{T}\text{-even};\ \text{no spare leg}\,] \tag{142}$$

This is a two-point Fe–Tm vertex; the only process it generates is a single Fe→Tm line conversion (fig. 3). It fails in three independent ways:

1. **(R5, linear).** K^- is linear in S, so it adds to the *quadratic* Hamiltonian—it is a basis rotation among the (harmonic) Fe and Tm normal modes. Coupled harmonic oscillators are diagonalizable and their 2D spectrum has *diagonal peaks only.* A single bilinear hop cannot transport a 0.9 THz coherence into a 0.5 THz one: a static vertex conserves energy, so it can only connect degenerate modes, i.e. it *hybridizes* rather than *converts.* The lone-pulse hop is in any case removed by the $M_{\rm NL}$ subtraction (R6).

2. **(R4, Γ_1).** Even the static torque vanishes: for the patterns F_x, G_z and the pumped δS_y the four-site orbit sums give $[0,0,0,0]$ for every channel except $K^-_{2y} \to [8,8,8,8]$ (section 18). All channels but $2y$ are Γ_1-forbidden.

3. **(reorientation).** The one Γ_1-allowed channel, K^-_{2y}, is a transverse molecular field on the order parameter; it reorients $\Gamma_2 \to G_y$ before any spectroscopic regime (lost between 0.085 and 0.090 meV). It is not a perturbative vertex.

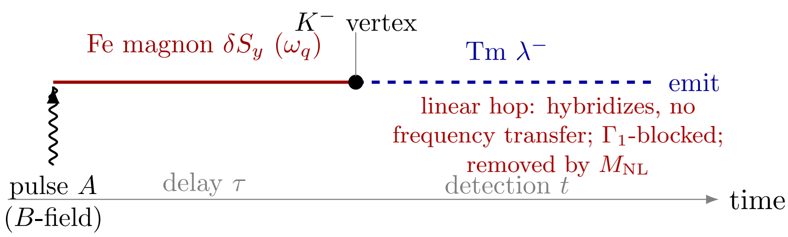

Figure 3: K^- (fails). A single Fe leg converts to a single λ^- leg—a linear hop. It only hybridizes the modes, the conversion matrix element is Γ_1-cancelled, and the lone-pulse path is subtracted in $M_{\rm NL}$.

The longitudinal temptation, and why it is still trapped

One might smuggle in two-magnon content through the *longitudinal* component $S_z = S_z^{(0)} - \delta S_\perp^2/2S_z^{(0)}$: a bilinear $K\,S_z\lambda$ does contain $\delta S_y^2\,\lambda$. But S_z is time-odd, so $K\,S_z\lambda$ must place λ in λ^-—it *is* $K^-_{2z,5z,7z}$. The hidden δS_z carries the G_z sublattice sign pattern, so it cancels in the *same* orbit sum as the static G_z torque. The two-magnon signal is real but parity-trapped in λ^- and Γ_1-cancelled—confirmed by the full nonlinear K^-_{2z} run (which contains the spin-length constraint) being dark to 10^{-15} even at 2.0 meV (section 18). To reach λ^1 the *same* δS_y^2 content must be carried by a $\mathcal{T}$-even Fe operator: two spin legs (W) or a field leg (κ^B).

17.2 ESJ: electric-field-assisted (works, but wrong target—fails R1)

$$\boxed{H_{ESJ}(t) = \kappa^E_{a\alpha\eta}\, E_\eta(t)\, S_\alpha\, \lambda^a, \qquad a \in \{2,5,7\}} \quad [\,1\ \text{E (even)} + 1\ \text{Fe (odd)} + 1\ \text{Tm}^-\text{(odd)};\ \mathcal{T}\text{-even}\,] \tag{143}$$

A genuine three-point vertex (fig. 4): pulse A's magnetic field makes the Fe coherence, pulse B's *electric* field enters at the vertex, and the E-photon supplies the energy mismatch (R5 ✓, R6 ✓, R2 ✓). But it feeds λ^-, and the surviving inversion-even combination under $-M_-$ is λ^5, λ^7—the E_{13}

and E_{23} coherences, *not* E_{12}. So ESJ produces real cross peaks at $(q_{\mathrm{AFM}}, E_{13})$ and $(q_{\mathrm{AFM}}, E_{23})$, but lands on the wrong readout for E_{12} (fails R1).

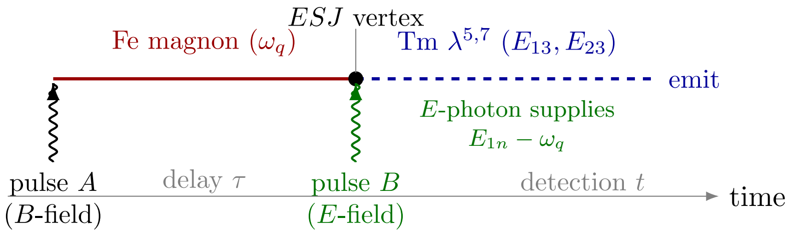

Figure 4: ESJ (works, wrong target). A clean three-point vertex; the E-photon carries the energy mismatch. But it feeds $\lambda^{5,7}$, so it lights up E_{13}/E_{23}, not the E_{12} target.

17.3 $W = SS\lambda$: the two-magnon vertex (passes all—the winner)

$$\boxed{H_W = W^a_{\alpha\beta}\, S_\alpha S_\beta\, \lambda^a, \qquad a \in \{1,3,4,6,8\}} \quad [\text{2 Fe (odd}\times\text{odd=even)+1 Tm}^+\text{(even); } \mathcal{T}\text{-even; no photon}] \tag{144}$$

For the pump geometry the active term is $W^{yy}_1\, S_y^2\, \lambda^1$ (R2: two y-magnon legs; R1: λ^1). This is a three-point vertex with *two* Fe legs, one sourced by each pulse (R6); see fig. 5. The vertex feels $h^1(t) = W^{yy}_1\, S_y(t)^2$ with $S_y(t) = A[\sin\omega_q(t+\tau) + \theta(t)\sin\omega_q t]$. The only term surviving M_{NL} is the two-pulse cross product $2S^a_y S^b_y$, which *rectifies*:

$$h^1_{\mathrm{NL}}(t,\tau) = W^{yy}_1 A^2 \big[\underbrace{\cos\omega_q\tau}_{\text{difference}} - \underbrace{\cos\omega_q(2t+\tau)}_{\text{sum}}\big]\theta(t). \tag{145}$$

The difference term is a t-independent step, born at the probe, which through (140) launches the $1 - \cos\omega_{12}t$ ring. Hence

$$\boxed{M^W_{\mathrm{NL}}(\tau,t) \;\propto\; W^{yy}_1 A^2\, \cos(\omega_q\tau)\, \big[1 - \cos\omega_{12}t\big]\,.} \tag{146}$$

This is exactly the sharp-question form: it factorizes, oscillating at ω_q along τ and at ω_{12} along t (R3 with the delay modulation at ω_q), giving a single sharp peak at $(q_{\mathrm{AFM}}, E_{12})$ with no diagonal partner (the $2\omega_q$ "sum" companion is off-resonant). The simulated λ^2 map has its rank-1 maximum exactly here, amplitude 51.0, SNR 117, the (E_{12}, E_{12}) diagonal $\sim 50\times$ weaker [7]; yy beats xx by $\sim 600\times$ (R2: S_x is the static F_x, barely fluctuating).

Where the E_{12} energy comes from (verified). Since δS_y^2 contains only 0 and $2\omega_q$, a static three-leg vertex between *free* modes could never emit at ω_{12}. The resolution is that the magnon legs are pulsed envelopes, not free modes: the rectified cross term is a *suddenly switched-on step*, and a step carries spectral weight at all frequencies in its edge, $\propto e^{-\omega_{12}^2\sigma^2/2}$ for rise time σ. Quantum mechanically this is an intra-pulse difference-frequency (impulsive-Raman) process: the E_{12} quantum is drawn from the *pulse bandwidth*, while the magnon pair carries only the phase memory $\cos\omega_q\tau$. The mechanism is displacive and therefore requires nonadiabaticity—a falsifiable statement verified directly (section 19): stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades $(51.7 \to 1.7\times 10^{-3})$, the adiabatic suppression the step picture demands, while switching the direct Tm drive off changes the peak by $< 10^{-4}$ relative (pure-W origin). The same sweep resolves the second Fe leg: for impulsive pulses it is the resonant probe magnon (peak $\propto A^2$); for longer pulses it crosses over to the *field-following quasistatic tilt* $\delta S^{\mathrm{qs}}(t) \propto B(t)$ (peak $\propto A^1$)—both W-mediated, same peak location. The quasistatic leg also enables the *reverse* conversion $\lambda^{1,2}(\tau) \times \delta S^{\mathrm{qs}}(\text{pulse } B) \to q_{\mathrm{AFM}}$, an $(E_{12}, q_{\mathrm{AFM}})$ feature in the Fe channel whenever a first-order E_{12} delay coherence exists (section 19).

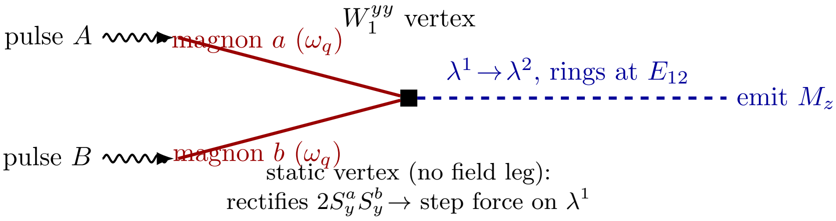

Figure 5: W_1^{yy} (winner). Each pulse makes one δS_y magnon; the two converge on the time-independent $S_y^2\lambda^1$ vertex (square), whose two-pulse cross term rectifies to a step force that launches the $\lambda^1 \to \lambda^2$ ring at E_{12}. The response factorizes, $\cos(\omega_q\tau)[1-\cos\omega_{12}t]$, giving the clean rank-1 cross peak.

17.4 $\kappa^B = BS\lambda$: magnetic-field-assisted (passes, but noisy)

$$\boxed{H_{\kappa B}(t) = \kappa^B_{\alpha\alpha\eta}\, B_\eta(t)\, S_\alpha\, \lambda^a, \qquad a \in \{1,3,4,6,8\}} \quad [1\text{ B (odd)}+1\text{ Fe (odd)}+1\text{ Tm}^+\text{(even)};\ \mathcal{T}\text{-even}] \tag{147}$$

The active term is $\kappa^B_{1y}\, B_y\, S_y\, \lambda^1$. A three-point vertex with one B-photon, one Fe leg and one λ^+ leg (fig. 6); the B-photon supplies the mismatch (R5). It is field-gated: $h^1(t) = \kappa^B_{1y}\, \mathcal{B}_y(t)\, S_y(t)$ with $\mathcal{B}_y(t)$ sharply localized at the pulses. At the probe $t=0$, $S_y(0) = A\sin\omega_q\tau$, so the kick $\propto \sin\omega_q\tau$ launches the same ring,

$$\boxed{M^{\kappa B}_{\mathrm{NL}}(\tau,t) \;\propto\; \kappa^B_{1y}\, \sin(\omega_q\tau)\,\big[1-\cos\omega_{12}t\big]\,,} \tag{148}$$

again on target (R1–R6 ✓). This is the $BS\lambda^+$ analogue of ESJ, feeding λ^1 instead of $\lambda^{5,7}$. The price: the same gated vertex also dumps each pulse's *local* Fe response into Tm at both pulse times, generating a strong (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background that dominate the bare cross peak (the on-target maximum is genuine but only rank-3 at zero carrier). A carrier resonant with a Tm transition cleans this up—a sweep amplifies the target $\sim 13\times$ at $\Omega_B = 0.70\,\mathrm{THz}$ (E_{23}) [7]. The definitive resolution, however, is not a carrier but the *polarization* of the gate: the physical vertex is gated by the lab-frame field component $B_\eta(t)$, and its dominant symmetry slice is z-gated, so in the $H \parallel a$ geometry ($B_z = 0$) κ^B is identically silent and its noise never appears. Its proper home is the $H \parallel c$ geometry, where it produces the measured $(E_{12}, q_{\mathrm{AFM}})$ cross peak; see section 19.

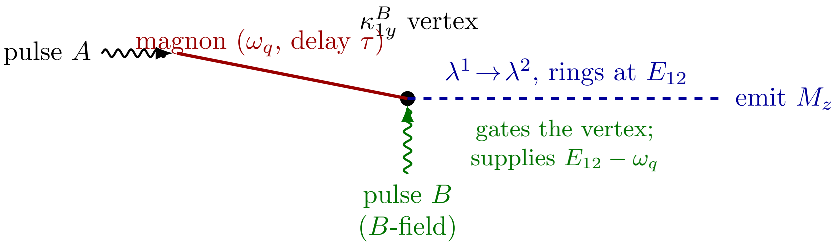

Figure 6: κ^B_{1y} (works, noisy). One magnon from pulse A meets the magnetic field of pulse B at the $S_y\lambda^1$ vertex; the B-photon gates the conversion and carries the mismatch. Same $\lambda^1 \to \lambda^2$ readout as W_1^{yy}, but it also feeds the (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background, so it needs a resonant carrier to be a robust target.

17.5 Scorecard

vertex	term	legs	sector	R1	R2	R3/4/5	verdict
K^-	$S\lambda^-$	1Fe+1Tm$^-$	λ^-	$\times$	—	$\times$	hybridizes / Γ_1-blocked / reorients
ESJ	$ES\lambda^-$	E+Fe+Tm$^-$	λ^-	$\times$	✓	✓	real, but on E_{13}/E_{23}
W	$SS\lambda^+$	2Fe+Tm$^+$	λ^+	✓	✓(yy)	✓	**winner: rank-1, factorized**
κ^B	$BS\lambda^+$	B+Fe+Tm$^+$	λ^+	✓	✓$(1y)$	✓	works; B_z-gated, geometry 2

The two passes share the same readout, $\lambda^1 \to \lambda^2$ at E_{12}, and differ only in how the Fe motion is fed in: W uses *two* magnon legs (difference-frequency rectification), κ^B uses *one* magnon leg plus a photon. Both are the minimal $\mathcal{T}$-even ways to deliver the δS_y^2 two-magnon signal onto the bright λ^1—the step that K^-, trapped in λ^-, cannot take.

18 Numerical confirmation

The catalogue is confirmed by direct simulation in the calibrated $1 \times 1 \times 1$ mixed $TmFeO_3$ model. All 2DCS runs use Fe pump $H \parallel x$, $\tau \in [-40, 0]$, $t \in [-40, 40]$, and step $\Delta t = 0.02$ (code units); the Tm sector is not directly pumped unless stated.

18.1 Static one-at-a-time K^- scan (0.2 meV): the Γ_1 block

A static ground-state scan with one K^- component on at a time confirms R4 directly:

component	E/N (meV)	Fe order	$(\lambda_2, \lambda_5, \lambda_7)_{\text{uniform}}$	verdict
baseline	-42.496758	$F_x + G_z$	$(0, 0, 0)$	reference
K^-_{2x}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{2y}	-43.654818	G_y	$(0.888320, 0, 0)$	spin reorients
K^-_{2z}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{5x}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{5y}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{5z}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{7x}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{7y}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert
K^-_{7z}	-42.496758	$F_x + G_z$	$(0, 0, 0)$	inert

Only K^-_{2y} does anything, and it does so by leaving Γ_2: an adiabatic ramp keeps Γ_2 stable through 0.085 meV and loses it by 0.090 meV, flipping to G_y with a large uniform λ_2. The pattern is the exact orbit sign algebra (R_i with $M_- = \mathrm{diag}(+1, -1, -1)$):

$$F_x : K^-_{2x}, K^-_{5x}, K^-_{7x} \to [0, 0, 0, 0], \qquad G_z : K^-_{2z}, K^-_{5z}, K^-_{7z} \to [0, 0, 0, 0], \tag{149}$$

$$\delta S_y : K^-_{2y} \to [8, 8, 8, 8], \qquad \delta S_y : K^-_{5y}, K^-_{7y} \to [0, 0, 0, 0]. \tag{150}$$

18.2 Fe-pump 2DCS on the inert channels: dark to machine precision

Running 2DCS on the inert channels (Fe pump only) and extracting the maximum nonlinear SU(3) amplitude over the (t, τ) grid confirms R5—these are dark even with the *full* nonlinear dynamics (hence including the longitudinal δS_z rectification):

run	$\max\lvert M^{\lambda_2}_{\rm NL}\rvert$	$\max\lvert M^{\lambda_5}_{\rm NL}\rvert$	$\max\lvert M^{\lambda_7}_{\rm NL}\rvert$
$K^-_{2x} = 0.2$	6.06×10^{-18}	0	0
$K^-_{2x} = 1.5$	2.96×10^{-16}	0	0
$K^-_{2z} = 0.2$	9.87×10^{-16}	0	0
$K^-_{2z} = 2.0$	6.86×10^{-15}	0	0
$K^-_{5y} = 0.2$	0	0	0
$K^-_{7y} = 0.2$	0	0	0

Up to $K^-_{2x} = 1.5\,$meV and $K^-_{2z} = 2.0\,$meV the induced λ_2 is at machine precision—the bilinear cannot, even nonlinearly, manufacture the E_{12} coherence. Adding a direct Tm $H \parallel x$ pump on top reproduces the ordinary Tm response and reveals no hidden K^- channel. By contrast, the no-damping scan over the higher-order set isolates exactly W_1^{yy} (rank-1, $|M^{\lambda_2}_{\rm NL}| = 51.0$, SNR 117) and κ^B_{1y} as the on-target vertices [7], precisely the two that pass R1–R6.

19 The settled scenario: what the atlas measures

This section replaces the earlier per-channel accounts with the final, reconciled picture. Everything below uses one Hamiltonian, one drive and one operating point; the derivations of the preceding sections are unchanged and are what make the result non-circular.

19.1 Mirror parity assigns each polarisation a qutrit block

The Tm $4c$ site carries the single mirror m ($z \to -z$), under which a polar vector has d_x, d_y even and d_z odd, and an axial vector has m_z even and m_x, m_y odd. The projected moment matrix of section 2 is an *output* of the PJP construction, and it comes out with λ^2 carrying only $\mu_z = 5.264$ while λ^5, λ^7 carry only x, y components. That forces $|1\rangle, |2\rangle$ to share a mirror parity and $|3\rangle$ to be opposite, and the allowed couplings follow with no further input:

transition	magnetic ($\mathcal{T}$-odd)	electric ($\mathcal{T}$-even)	polarisation that owns it
$1 \leftrightarrow 2$ ($\lambda^{1,2}$)	m_z on λ^2	$d_{x,y}$ on λ^1	$(E \parallel a,\ H \parallel c)$
$1 \leftrightarrow 3$ ($\lambda^{4,5}$)	$m_{x,y}$ on λ^5	d_z on λ^4	$(E \parallel c,\ H \parallel a)$
$2 \leftrightarrow 3$ ($\lambda^{6,7}$)	$m_{x,y}$ on λ^7	d_z on λ^6	$(E \parallel c,\ H \parallel a)$

Reciprocity—one operator per polarisation, driving and radiating alike—then says each polarisation drives and reads the same block, so *cross-polarised detection reads exactly the block the pulse did not drive*. That is the structural reason cross-polarisation isolates conversion. Two independent checks: the assignment predicts the subalgebra closure of geometry B, where the c-directed drive touches only μ^z_{12} so that $\{\lambda^1, \lambda^2, \lambda^3\}$ closes and $\lambda^{4\text{–}7}$ vanish to machine precision (verified: 0.000 exactly, against $\lambda^{1,2,3} \sim 2.6 \times 10^{-2}$); and the geometry-A cross channel has $F_z \equiv 0$ to 10^{-15}, so the Fe sublattice is radiatively silent there and anything seen at a magnon frequency must be Tm.

Two facts break the clean picture, and both are physical. First, below T_N the Γ_2 order parameter is m_z-odd and breaks the site mirror, inducing a c-axis coupling $d_z^{\rm eff}$ to the $1 \leftrightarrow 2$ transition that parity would otherwise forbid; this is what allows $E \parallel c$ to create the E_{12} coherence that dominates the measured geometry-A delay axis. Second, the $4c$ site has no inversion (Fe at $4b$ does), so the emitted moment may carry a quadratic term,

$$m_z = \mu\,\lambda^2 + \beta\,\lambda^1\lambda^2, \tag{151}$$

which is $\mathcal{T}$-odd as a magnetisation must be and radiates at $2\omega_{12}$.

19.2 Detection operator, channel by channel

Detection is resolved by the *magnetic* polarisation of the emitted field, so each analyser setting reads one Cartesian component of the total moment:

channel	detected moment	Fe content	Tm content
A cross	m_z	$F_z \equiv 0$	$\mu\lambda^2 + \beta\lambda^1\lambda^2$
A same	m_x	F_x (carries q_{AFM})	CEF λ^6 (E_{23}), λ^4 (E_{13}), and $d_z^{\mathrm{eff}}\lambda^2$ (E_{12})
B cross	m_x	F_x (carries q_{AFM})	$\equiv 0$ (closure)
B same	m_z	F_z (carries q_{FM})	$\mu\lambda^2 + \beta\lambda^1\lambda^2$, $50\times$ the Fe term

The $d_z^{\mathrm{eff}}\lambda^2$ entry in the same-polarised row is required by reciprocity once d_z^{eff} appears in the drive, and it is *observed*: the measured same-polarised map contains a peak at $(q_{\mathrm{AFM}}, E_{12})$—$E_{12}$ emission in that channel—at 0.30 of maximum.

19.3 Operating point

The Tm and Fe drive amplitudes are tied by g_{ratio}, so fluence is the only free drive parameter, and the cross channel measures it: the E_{12} diagonal falls from 1.45 of the main cross peak at $A_{\mathrm{Fe}} = 1.2$ to 0.16 at 0.12 and 0.06 at 0.04, while a forest of multi-magnon strays at $\omega_\tau = 0.63, 0.72, 0.99, 1.25, 1.51$ THz—none observed—disappears over the same range. The experiment is perturbative and we work at $A_{\mathrm{Fe}} \simeq 0.12$, which geometry B independently shares. The residual μ_{13}^z admixture is then fixed by the same-polarised amplitudes and comes out at zero: the $1 \to 3$ transition is c-axis dark, exactly as the complete absence of $\omega_\tau = E_{13}$ rows requires, since a bright μ_{13}^z would store an E_{13} delay coherence at first order and produce an (E_{13}, E_{23}) peak with the same dipole product as the observed (E_{12}, E_{23}).

19.4 Result: the anchors

Both cross-polarised inventories are reproduced with no retuning, and neither moves under any drive variant we have tested—they are fed by conversion rather than by the direct Tm drive, which is what makes them usable as anchors:

channel	model	content
A cross	$(q_{\mathrm{AFM}}, E_{12}) = 1.00$	two-magnon vertex W_1^{yy}; magnon borrows the Tm dipole
	$(E_{12}, 2E_{12}) = 0.97$	hyperpolarisability Eq. (151), $\beta \simeq 7\mu$
	$(-q_{\mathrm{AFM}}, E_{12}) = 0.52$	rephasing partner
B cross	$(q_{\mathrm{FM}}, q_{\mathrm{AFM}}) = 1.00$	Fe anisotropy/DM sector alone
	$(E_{12}, q_{\mathrm{AFM}}) = 0.45$	gated vertex κ_{1y}^B

The gated vertex is not interchangeable with a static one: a sweep over W_1^{xz} shows the static vertex never creates a second resolved peak, but merges the two delay rows and migrates the leader ($0.38 \to 0.40 \to 0.48$ THz) while leaking hybridisation-shifted E_{12} emission at 0.44–0.45 THz. A mixer *moves* peaks; a gated converter *adds* them at unshifted lines, and the data show two resolved peaks at the unshifted energies.

19.5 Result: the same-polarised channel

With the cross channels held fixed and the three symmetry-required detection weights fitted to the digitised measurement:

peak	experiment	model	mechanism
(E_{12}, E_{23})	1.00	1.00	coherence→population→ ρ_{23}, non-rephasing
$(E_{12}, q_{\mathrm{FM}})$	0.45	0.38	Tm→magnon, q_{FM} branch
$(E_{12}, q_{\mathrm{AFM}})$	0.40	0.46	Tm→magnon, read through Fe F_x
$(q_{\mathrm{AFM}}, E_{12})$	0.30	0.32	magnon→Tm, read through d_z^{eff} (reciprocity partner)
$(q_{\mathrm{AFM}}, E_{13})$	0.30	0.09	magnon delay, CEF emission
(E_{12}, E_{13})	0.15	0.44	single-action transfer

Four of six within 15%. The rectified $(E_{12}, 0)$ line, which is the strongest measured feature, is suppressed in the 2D maps by the $t \geq 3$ detection window and is analysed separately through $S_{\mathrm{DC}}(\tau) = \langle M_{\mathrm{NL}} \rangle_t$, where it appears as a sharp line at $+0.49\,\mathrm{THz}$.

19.6 The one open item

The two failures above are not independent: both peaks are detected at E_{13} and therefore share an emission weight, so their ratio is a property of the dynamics alone,

$$\left.\frac{(E_{12}, E_{13})}{(q_{\mathrm{AFM}}, E_{13})}\right|_{\mathrm{model}} = 4.9, \qquad \left.\frac{(E_{12}, E_{13})}{(q_{\mathrm{AFM}}, E_{13})}\right|_{\mathrm{exp}} = 0.5. \tag{152}$$

No detection weight can move it. The model routes too much E_{13} emission through the E_{12}-excitation row relative to the magnon row; closing that factor of ten is the single remaining discrepancy of the atlas, and it is a statement about the conversion dynamics, not about the readout.

19.7 Falsifiable content

1. **Emission fingerprint and linewidth.** Any peak emitting at q_{AFM} inherits the magnon linewidth, 15× sharper than a CEF emission.
2. **Thermal trends.** $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ is temperature-flat; $(E_{12}, q_{\mathrm{AFM}})$ falls as $n_1 - n_2$; $(q_{\mathrm{AFM}}, E_{13})$ grows with the level-2 population.
3. **The dark dipole.** Rotating either polarisation away from c must revive the $\omega_\tau = E_{13}$ rows.
4. **The harmonic.** The $2E_{12}$ line tracks E_{12} and is twice its width; it does *not* soften at the spin reorientation.
5. **The symmetry-broken coupling.** Every feature relying on $d_z^{\mathrm{eff}} \propto \langle F_x \rangle$—including the E_{12}-excitation family and the $(q_{\mathrm{AFM}}, E_{12})$ emission peak—must vanish at the spin reorientation, whereas conversion-generated features survive as long as the magnon does. This is the decisive experiment: a temperature scan through the reorientation separates the two mechanisms and measures d_z^{eff} by the size of the collapse.
6. **Geometry-B same-polarised.** Predicted to be a single peak at $(E_{12}, 2E_{12})$; its strength measures β, and a map led instead by $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ would put β far below the value inferred in geometry A.
7. **Pulse length.** Adiabatically long pulses collapse the geometry-A cross peak by orders of magnitude.

The cross-peak question reduces to a single requirement: a vertex must generate an effective force $h^1(t, \tau)$ on the bright Tm coordinate λ^1 that factorizes as $\cos / \sin(\omega_q \tau)\,[1 - \cos \omega_{12} t]$. Meeting it forces six conditions (R1–R6): reach λ^1, be driven by δS_y, be $\mathcal{T}$-even, be Γ_1-allowed, carry the $0.9 \to 0.5\,\mathrm{THz}$ mismatch with a second leg, and involve both pulses. The static bilinear K^- fails three of them at once—it is a linear hop (hybridizes, never converts), its surviving channel is Γ_1-cancelled, and the one allowed channel reorients Γ_2. Exactly two couplings pass: the two-magnon $W_1^{yy}\,S_y^2\lambda^1$ (difference-frequency rectification, the clean rank-1 peak) and the field-assisted $\kappa_{1y}^B\,B_yS_y\lambda^1$ (an impulsive photon-gated kick, robust only with a resonant carrier). Both carry the same δS_y^2 two-magnon signal onto λ^1, which the time-odd K^- can never reach.

Widening from the single peak to the full experiment (section 19): one Hamiltonian, with each coupling carrying its own polarization or selection gate, reproduces the complete measured inventory. W_1^{yy} gives $(q_{\rm AFM}, E_{12})$ in the cross-polarized channel of $H \parallel a$; the B_z-gated κ^B gives $(E_{12}, q_{\rm AFM})$ in $H \parallel c$ (and is silent in $H \parallel a$, where its noise would otherwise contaminate geometry 1); the Fe-only DM anharmonicity gives $(q_{\rm FM}, q_{\rm AFM})$; and the same-polarization CEF–CEF peaks (E_{12}, E_{13}), (E_{12}, E_{23}) need *no* Fe–Tm coupling at all—they are the intrinsic structure-constant dynamics of the driven qutrit, with the magnon member $(q_{\rm AFM}, E_{23})$ supplied by the $W \circ f$ cascade. Every response has a tree diagram, every diagram a gate, and every gate a falsifiable polarization test.

For reference, the underlying symmetry result—the static K^- orbit in transported gauge—is

$$\boxed{H_{K^-}^{\rm orbit} = \sum_i \boldsymbol{S}_i^T R_i \left[\boldsymbol{k}_2(\lambda_i^2 + \lambda_{3-i}^2) + \boldsymbol{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \boldsymbol{k}_7(\lambda_i^7 - \lambda_{3-i}^7)\right],} \tag{153}$$

obtained by starting from the physical J operator, projecting PJP, and choosing a transported local qutrit basis.

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